A-A-50968
11 June 1984

COMMERCIAL ITEM DESCRIPTION

CHUCKS, DRILL, KEYLESS TYPE, THREE JAW, BALL BEARINGS

The General Services Administration has authorized the use of this commercial item description in preference to the Type II chucks covered by Federal specification GGG-C-350.

This description covers keyless type, three jaw chucks which are designed to hold straight shank tools within the capacity of the chuck. These chucks are used on high accuracy drill presses, jig borers, and milling machines where frequent drill changes are required.

Salient characteristics.

Classification. The drill chucks covered by this description shall be of the following types and of the sizes specified in table I or table II. The type and size chuck required for a specific requirement shall be as specified in the contract or order.

Type I - Tapered mount
Type II - Threaded mount

Sizes and capacities. The type I and type II drill chuck shall meet the size and capacity requirements of table I or table II.

Design. The drill chucks covered by this CID shall be of the latest design, easy to operate and entirely adequate for gripping various sizes of straight shank tools.

Hardened jaws. The gripping surfaces of the jaws shall have a Rockwell C scale hardness value of not less than 52.

Hardened bore. The taper mount bore (Type I) shall be hardened in accordance with the manufacturer's standard practice and shall conform to the dimensional requirements of table I.

Threads. Type II (thread mount) drill chucks shall have right-hand threads in the mount hole which shall meet the requirements of table II for the chuck size specified. All threads shall conform to FED-STD-H28.

Bearings. Thrust bearings used in the chuck shall be of a type and class to provide the necessary accuracy required.

FSC 3460

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TABLE I. Sizes and capacities of type I chucks.

Size number	Nominal capacity, inches	Jacobs taper	Chuck dimensions, inches	
			Length with jaw closed not less than	Diameter not less than
1	0-1/16	0	1-15/32	3/4
2	0-1/8	0	1-7/8	15/16
3	0-1/8	1	1-7/8	15/16
4	0-5/32	0	1-23/32	1
5	0-5/32	0	2-1/32	1-1/64
6	0-5/32	1	2-1/32	1-1/64
7	0-3/16	1	2-7/16	1-3/16
8	0-1/4	1	2-1/16	1-9/64
9	0-1/4	1	2-35/64	1-1/4
10	0-5/16	1	2-19/64	1-19/64
11	0-5/16	1	2-7/8	1-3/8
12	0-5/16	2 short	2-23/64	1-31/64
13	0-5/16	2 short	3-1/16	1-1/2
14	0-3/8	2	3-31/32	1-21/32
15	0-3/8	2	3-1/2	1-37/64
16	0-3/8	33	3-1/32	1-21/32
17	0-3/8	33	3-17/32	1-9/16
18	0-1/2	6	4-3/64	1-31/32
19	1/32-1/2	2	3-11/32	2-1/128
20	1/32-1/2	2	3-31/32	1-13/16
21	1/32-1/2	2	4-1/16	1-31/32
22	1/32-1/2	6	3-3/16	1-13/16
23	1/32-1/2	6	4-1/16	1-31/32
24	1/32-1/2	33	3-3/16	1-13/16
25	1/32-1/2	33	4-1/16	1-31/32
26	1/8-5/8	6	4-3/16	2-1/64

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TABLE II. Sizes and capacities of type II chucks.

Size number	Nominal capacity, inches	Nominal diameter and threads	Chuck dimensions, inches	
			Length with jaw closed, not less than	Diameter not less than
1	0-1/8	5/16-24	1-7/8	15/16
2	0-3/16	5/16-24	2-7/16	1-3/16
3	0-1/4	5/16-24	2-11/16	1-11/32
4	0-1/4	3/8-24	2-1/16	1-9/64
5	0-1/4	3/8-24	2-21/32	1-1/4
6	0-1/4	1/2-20	2-23/64	1-19/64
7	0-1/4	1/2-20	2-11/16	1-11/32
8	0-5/16	3/8-24	2-23/64	1-19/64
9	0-5/16	3/8-24	2-63/64	1-3/8
10	0-5/16	1/2-20	2-23/64	1-19/64
11	0-5/16	1/2-20	2-63/64	1-31/64
12	0-3/8	1/2-20	3-19/32	1-11/16
13	0-3/8	5/8-16	3-19/32	1-11/16
14	1/32-1/2	1/2-20	2-7/8	1-21/32
15	1/32-1/2	1/2-20	3-19/32	1-37/64
16	1/32-1/2	1/2-20	4-1/16	1-31/32
17	1/32-1/2	5/8-16	3-27/64	2-1/128
18	1/32-1/2	5/8-16	4-1/16	1-31/32
19	1/32-1/2	5/8-16	4-51/64	1-51/64
20	1/32-1/2	3/4-16	4-1/16	1-31/32
21	1/16-3/8	3/8-24	2-63/64	1-31/64
22	1/16-3/8	1/2-20	2-63/64	1-31/64
23	1/16-1/2	1/2-20	3-19/32	1-37/64
24	1/8-5/8	5/8-16	4-5/64	1-51/64
25	1/8-5/8	3/4-16	4-9/32	2-7/32

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Markings. Each drill chuck shall be marked in a permanent and legible manner with the name of the manufacturer and chuck capacity.

Sleeve surface. The outside surface of the chuck sleeve shall be knurled, fluted, or serrated to provide a non slip surface.

Gripping power. The chuck shall be capable of gripping the straight shank of a tool firmly enough to prevent the tool from turning in the chuck jaws under operating conditions. The gripping power shall automatically increase when the load upon the tool increases.

Finishes. Each drill chuck shall be free from any imperfections or defects which may impair serviceability, durability or appearance.

Measurement system. All dimensions in this commercial item description are given in the English system of measurements. These dimensions can be converted to Metric system of equivalents by the use of ANSI/IEEE 268 for Department of Defense use of FED-STD-376 for Civil Agency use.

Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the CID where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

Quality conformance inspection. The supplier shall randomly select the number of samples specified in table III that is applicable to the quantity of items being purchased. Each sample shall be visually inspected, measured, and tested, by the supplier, to determine conformance with all requirements specified herein. Any sample failing to meet all requirements specified herein shall be considered a defective item. If the number of defective items equals, or is less than the quantity specified under "Acceptance, number defective" column of table III, the supplier shall replace the defective samples with items that meet all of the requirements specified herein and the entire lot shall be considered acceptable for shipment. If the number of defective samples equals, or exceeds, the quantity specified in the "Rejection, number defective" column of table III, the entire lot shall be considered defective and unsuitable for acceptance.

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TABLE III. Sampling plan.

Quantity of items being purchased	Sample size	Acceptance number defective	Rejection, number defective
2 to 8	2	0	1
9 to 15	3	0	1
16 to 25	5	0	1
26 to 50	8	1	2
51 to 90	13	1	2
91 to 150	20	2	3
151 to 280	32	3	4
281 to 500	50	5	6

Hardness test. A hardness test shall be performed on each sample drill chuck selected in accordance with table III. If the test samples fail the applicable hardness test specified herein, the heat treat lot from which the samples were selected shall be rejected.

Smoothness of operation. Each of the sample drill chucks selected in accordance with table III shall operate smoothly, when revolved by hand, from the closed position to the largest capacity without binding or having excessive play between parts at any point.

Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified in the contract or order, preservation, packaging, packing, labeling and marking shall be in accordance with ASTM D-3951. When specified in the contract or order, preservation, packaging, packing, labeling and marking shall be in accordance with Federal Specification PPP-T-1150, level as specified.

Notes. Purchasers shall specify quantity, type and size chuck required for a specific requirement.

FED-STD-376, "Preferred Metric Units For General Use By The Federal Government" and FED-STD-H28, "Screw-Thread Standards for Federal Services" should be obtained from the contracting activity or as directed by the contracting officer.

American National Standards Institute ANSI/IEEE 268 "Metric Practice" is available from the American National Standards Institute, 1430 Broadway, New York, NY 10018.

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MILITARY INTERESTS:

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Air Force - 99

Review Activities

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CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

PREPARING ACTIVITY:

DLA - IP

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